REVISION WORKSHEET

UNITED ARAB EMIRATES
MINISTRY OF EDUCATION

الإمارات العربية المتحدة
وزارة التربية والتعليم

AL KHALIDIYA BOYS SCHOOL

Cycle 3 (10048) | Dibba Al Hisn | Sharjah Eastern

Diffraction Practice Questions

Grade 10 Advanced | Module 17 Lesson 2 | Diffraction

Name: _____ Class: _____ Date: _____

Answer all 30 questions. Select one option for each MCQ. Show equations, working, units and explanations for written questions.

Useful equations and guidance: Single slit: $W \approx 2\lambda L/w$. Grating: $d \sin\theta = m\lambda$; $d = 1/(\text{line density})$. Use consistent units.

Term 1 2026–2027 | Textbook alignment: uploaded Grade 10 book, printed pages 50–57.

P01. Diffraction is most noticeable when an opening is:

A. Infinitely wide B. Much wider than every relevant scale
C. Comparable in size to the wavelength D. Made of a single specific colour

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P02. For a single slit, the central bright maximum is generally:

A. A perfectly dark line B. Infinitely narrow
C. Always identical to a double-slit fringe D. Wider than neighbouring bright maxima

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P03. Narrowing a single slit while keeping wavelength fixed makes the central maximum:

A. Wider B. Narrower
C. Disappear at all angles D. Have zero intensity necessarily

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P04. A diffraction grating contains:

A. Only one thick lens B. Many closely spaced parallel slits or grooves
C. Only one plane mirror D. No periodic structure

P05. A grating spectroscope separates light mainly by:

A. Changing every photon's frequency B. Absorbing only red light
C. Wavelength-dependent diffraction angles D. Making all wavelengths leave at one angle

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P06. A single slit has width 0.200 mm. With λ = 500 nm and screen distance 2 m, find the central bright-band width in millimetres. Use $W = 2\lambda L/w$.

A. 10 mm B. 20 mm
C. 5 mm D. -10 mm

P07. A single-slit pattern has central width 20 mm at screen distance 2 m with λ = 500 nm. Find the slit width in millimetres.

A. 0.2 mm B. 0.1 mm
C. 0.05 mm D. -0.1 mm

P08. A diffraction grating has spacing d = 2000 nm. For λ = 500 nm, calculate the angle of the first-order maximum.

A. 28.96 ° B. 7.239 °
C. 14.48 ° D. -14.48 °

P09. A grating has 300 lines per millimetre. Calculate its slit spacing in micrometres.

A. 6.667 µm B. 1.667 µm
C. -3.333 µm D. 3.333 µm

P10. A grating spacing is 2000 nm and the wavelength is 500 nm. Find the largest possible integer diffraction order.

A. 4 B. 8
C. 2 D. -4

P11. Define diffraction.

P12. What determines the single-slit central-maximum width in the small-angle approximation?

P13. Define diffraction order.

P14. Give an example of a reflection grating.

P15. Why are higher grating orders limited?

P16. A single slit has width 0.200 mm. With λ = 500 nm and screen distance 2.5 m, find the central bright-band width in millimetres. Use W = 2λL/w.

P17. A single slit has width 0.200 mm. With λ = 500 nm and screen distance 3 m, find the central bright-band width in millimetres. Use W = 2λL/w.

P18. A single-slit pattern has central width 30 mm at screen distance 3 m with λ = 500 nm. Find the slit width in millimetres.

P19. A single-slit pattern has central width 40 mm at screen distance 4 m with λ = 500 nm. Find the slit width in millimetres.

P20. A diffraction grating has spacing d = 2000 nm. For λ = 550 nm, calculate the angle of the first-order maximum.

P21. A diffraction grating has spacing d = 2000 nm. For λ = 600 nm, calculate the angle of the first-order maximum.

P22. A grating has 350 lines per millimetre. Calculate its slit spacing in micrometres.

P23. A grating has 400 lines per millimetre. Calculate its slit spacing in micrometres.

P24. A grating spacing is 2000 nm and the wavelength is 550 nm. Find the largest possible integer diffraction order.

P25. A grating spacing is 2000 nm and the wavelength is 600 nm. Find the largest possible integer diffraction order.

P26. Compare single-slit diffraction of red and blue light with the same slit and screen distance.

P27. If a single slit is widened to twice its initial width, predict the new central bright-band width.

P28. For a grating, why are red maxima at larger first-order angles than blue maxima?

P29. A student confuses grating line density with spacing. Explain the correct conversion.

P30. Describe how a grating can help identify an unknown gas-discharge source.